Non-Homomorphism Factors and the Concentration Parameter in Graph Homomorphisms

Mohammad Ali Rostami

March 26, 2026

Abstract

We study two measures associated with graph homomorphisms. The *non-homomorphism factor* $|G, H|$ is the minimum number of edges that must be removed from G to obtain a graph admitting a homomorphism to H. We prove $|K_n, K_m| = \binom{n}{2} - t(n, m)$ via Turán's theorem and derive a lower bound $|G, H| \geq |E(G)| - t(|V(G)|, \chi(H))$ for arbitrary G. We introduce the *concentration parameter* $\gamma(G, H)$, the minimum over all homomorphisms $f\colon G \to H$ of the maximum number of G-edges mapped to any single H-edge. We prove a multiplicativity inequality $\gamma(G, K) \leq \gamma(G, H) \cdot \gamma(H, K)$, a monotonicity property $\gamma(G, H_2) \leq \gamma(G, H_1)$ when $H_1 \subseteq H_2$, and the bound $|G, K| \leq \gamma(G, H) \cdot |H, K|$. Computational experiments on cycles, complete graphs, complete bipartite graphs, and the Petersen graph yield exact values of γ and reveal that the symmetric combination $m(G, H) = \max(|G, H|, |H, G|)$, previously claimed to be a pseudo-metric, does *not* in general satisfy the triangle inequality (108 counterexample triples found among complete graphs with at most 10 vertices). We also prove an exact formula for $\gamma(C_n, K_3)$ for all $n \geq 3$, verified computationally up to $n = 50$, and a general d-regularity theorem that unifies the cycle formula with the value $\gamma(\text{Petersen}, K_3) = 6$. Extended experiments on irregular source graphs (wheel, friendship, and book graphs), non-complete targets (odd cycles, the Petersen graph as target), and Kneser graphs demonstrate that γ exactly equals the pigeonhole bound for friendship graphs ($\gamma(F_k, K_3) = k$) but strictly exceeds it for wheel graphs and complete multipartite graphs, and that the parameter γ captures local congestion phenomena fundamentally different from the non-homomorphism factor.

1 Introduction

Throughout this paper all graphs are simple and undirected.

Definition 1.1 (Graph Homomorphism)**.** *A* homomorphism *from G to H is a function $f\colon V(G) \to V(H)$ such that $\{u, v\} \in E(G)$ implies $\{f(u), f(v)\} \in E(H)$. We write $G \to H$ when such a function exists.*

A homomorphism need not be injective or surjective. A bijection whose inverse is also a homomorphism is an *isomorphism*; an isomorphism from G to itself is an *automorphism*.

A proper k-colouring of G exists if and only if $G \to K_k$, giving:

Proposition 1.1 ([4])**.** *$G \to H \Rightarrow \chi(G) \leq \chi(H)$.*

Theorem 1.1 ([4])**.** *Any homomorphism $f\colon G \to H$ satisfies $d_H(f(u), f(v)) \leq d_G(u, v)$ for all $u, v \in V(G)$.*

Theorem 1.2 ([4])**.** *If H is vertex-transitive and $G \to H$ then $\dfrac{\alpha(G)}{|V(G)|} \geq \dfrac{\alpha(H)}{|V(H)|}$, where α denotes the independence number.*

1.1 Cores

Definition 1.2. *A graph is a* core *if it has no homomorphism to any proper subgraph of itself. Every graph G contains a unique (up to isomorphism) core subgraph, denoted* $\mathrm{core}(G)$*, such that $G \to \mathrm{core}(G)$ and $\mathrm{core}(G) \subseteq G$.*

A key consequence is that $K \to G$ if and only if $K \to \mathrm{core}(G)$. The Petersen graph is its own core, as it admits no homomorphism to any proper subgraph [4].

2 Non-Homomorphism Factors

Definition 2.1 (Non-Homomorphism Factor [1]). *$|G, H|$ is the minimum number of edges that must be removed from G to obtain a graph G' with $G' \to H$. If $G \to H$ then $|G, H| = 0$.*

Proposition 2.1 ([1]). *1. $H_1 \to H_2 \Rightarrow |G, H_1| \geq |G, H_2|$ for every graph G.*

2. $H_1 \leftrightarrow H_2 \Rightarrow |G, H_1| = |G, H_2|$ for every G.

3. If $f\colon G_1 \to G_2$ is edge-injective (distinct edges map to distinct edges) then $|G_1, H| \leq |G_2, H|$ for every H.

Proof. (1) If $G' \to H_1$ then $G' \to H_1 \to H_2$, so every graph witnessing $|G, H_1|$ also witnesses $|G, H_2|$. (2) Apply (1) in both directions. (3) Let $S \subseteq E(G_2)$ with $|S| = |G_2, H|$ such that $G_2 - S \to H$. Set $S' = \{e \in E(G_1) : f(e) \in S\}$. Since f is edge-injective, $|S'| \leq |S|$. The restricted map $f\colon G_1 - S' \to G_2 - S$ is still a homomorphism, so $G_1 - S' \to H$, giving $|G_1, H| \leq |S'| \leq |G_2, H|$. □

2.1 Complete Graphs and Turán's Theorem

Theorem 2.1 (Turán [4]). *Among all K_{m+1}-free graphs on n vertices the one with the most edges is the* Turán graph *$T(n, m)$, obtained by partitioning the n vertices into m classes of sizes as equal as possible and taking all edges between distinct classes; write $t(n, m) = |E(T(n, m))|$.*

Theorem 2.2. $|K_n, K_m| = \binom{n}{2} - t(n, m)$. *If $n = qm + r$ with $0 \leq r < m$, this equals $r\binom{q+1}{2} + (m - r)\binom{q}{2}$. In particular, $|K_{n+1}, K_n| = 1$ for all $n \geq 1$.*

Proof. A spanning subgraph G' of K_n satisfies $G' \to K_m$ if and only if $\chi(G') \leq m$. The maximum number of edges an m-colourable graph on n vertices can have is $t(n, m)$ (Turán's theorem), so removing the minimum number of edges to achieve m-colourability requires removing exactly $\binom{n}{2} - t(n, m)$ edges. The explicit formula follows from $t(n, m) = \binom{n}{2} - r\binom{q+1}{2} - (m - r)\binom{q}{2}$. For $|K_{n+1}, K_n|$: set $G = K_{n+1}$, $m = n$, so $q = 1$, $r = 1$, giving $1 \cdot \binom{2}{2} + (n - 1) \cdot \binom{1}{2} = 1$. □

Table 1: $|K_n, K_m|$ for $n = 2, \ldots, 7$ and $m = 2, \ldots, 6$. Values confirmed by exhaustive edge-removal search (Section 5) and by Theorem 2.2.

	K_2	K_3	K_4	K_5	K_6
K_2	0	0	0	0	0
K_3	1	0	0	0	0
K_4	2	1	0	0	0
K_5	4	2	1	0	0
K_6	6	3	2	1	0
K_7	9	5	3	2	1

2.2 A General Lower Bound

Theorem 2.3. *For any graphs G and H,*

$$|G, H| \;\geq\; \max\big(0,\, |E(G)| - t(|V(G)|, \chi(H))\big).$$

Proof. Any $G' \subseteq G$ with $G' \to H$ must satisfy $\chi(G') \leq \chi(H)$. By Turán's theorem, $|E(G')| \leq t(|V(G)|, \chi(H))$. Hence at least $|E(G)| - t(|V(G)|, \chi(H))$ edges must be removed from G. □

The bound is tight for $G = K_n$: it then equals $|K_n, K_m|$ from Theorem 2.2. For sparser G the bound may be 0, but then $|G, H|$ can be positive due to structural obstructions beyond edge density.

2.3 Cycles

Every cycle C_n satisfies $C_n \to K_3$: even cycles are bipartite, so $C_n \to K_2 \subset K_3$; odd cycles are 3-chromatic, so $C_n \to K_3$ directly. Hence $|C_n, K_3| = 0$ for all $n \geq 3$.

By contrast, $|C_n, K_2| = 0$ for even n and $|C_n, K_2| = 1$ for odd $n \geq 3$ (removing any edge produces a path, which is bipartite).

2.4 The Petersen Graph

The Petersen graph P has 10 vertices, 15 edges, chromatic number 3, and is its own core. Since $\chi(P) = 3$, $|P, K_m| = 0$ for all $m \geq 3$.

Observation 2.1. $|P, K_2| = 3$. *The lower bound from Theorem 2.3 gives* $|P, K_2| \geq |E(P)| - t(10, 2) = 15 - 25 = 0$, *which is not tight (the Petersen graph is far from bipartite). The value* 3 *is verified computationally in Section 5.*

3 The Concentration Parameter

Definition 3.1 (Concentration Parameter)**.** *Let G and H be graphs with $|E(G)| \geq 1$. For a homomorphism $f\colon G \to H$ and an edge $e \in E(H)$, the* load *of e under f is*

$$\mu(f, e) = \big|\{e' \in E(G) : \{f(u), f(v)\} = e \text{ for the endpoints } u, v \text{ of } e'\}\big|.$$

Define the max edge multiplicity $\mu(f) = \max_{e \in E(H)} \mu(f, e)$. *The* concentration parameter *is*

$$\gamma(G, H) = \min_{f\colon G \to H} \mu(f).$$

When $G \not\to H$ we set $\gamma(G, H) = +\infty$.

Remark 3.1. *Note that α denotes the independence number throughout; the symbol μ is used for edge multiplicities to avoid collision. A related per-homomorphism quantity M^σ appears in [2].*

Since the $|E(G)|$ pre-images are distributed among $|E(H)|$ images, the pigeonhole principle gives:

Proposition 3.1 (Pigeonhole lower bound)**.** *If $G \to H$ then $\gamma(G, H) \geq \lceil |E(G)|/|E(H)| \rceil$.*

This bound is not always achieved; Table 2 shows cases where the cycle structure forces a higher concentration.

Lemma 3.1. $\gamma(G, G) = 1$ *for any graph G with $|E(G)| \geq 1$.*

Proof. The identity $\mathrm{id}\colon G \to G$ satisfies $\mu(\mathrm{id}) = 1$ since each edge is its own unique pre-image. Since $\gamma \geq 1$ by definition (every edge must map somewhere), $\gamma(G, G) = 1$. □

Lemma 3.2. *If $G \to H$ and $|E(G)| \geq 1$ then $\gamma(G, H) \geq 1$.*

Proof. Every edge of G must map to an edge of H (since H has no loops), so some H-edge receives load at least 1. □

Lemma 3.3 (Multiplicativity)**.** *If $G \to H \to K$ then $\gamma(G, K) \leq \gamma(G, H) \cdot \gamma(H, K)$.*

Proof. Choose $f\colon G \to H$ with $\mu(f) = \gamma(G, H)$ and $g\colon H \to K$ with $\mu(g) = \gamma(H, K)$. Consider the composition $g \circ f\colon G \to K$. For any K-edge e, the G-edges mapping to e under $g \circ f$ are exactly those mapping under f to some H-edge in $g^{-1}(e)$. There are at most $\mu(g) = \gamma(H, K)$ such H-edges, and each carries at most $\mu(f) = \gamma(G, H)$ pre-images from G. Hence $\mu(g \circ f) \leq \gamma(G, H) \cdot \gamma(H, K)$, so $\gamma(G, K) \leq \mu(g \circ f) \leq \gamma(G, H) \cdot \gamma(H, K)$. □

Lemma 3.4 (Bound on non-homomorphism factor)**.** *If $G \to H$ then $|G, K| \leq \gamma(G, H) \cdot |H, K|$ for every graph K.*

Proof. Let $k = |H, K|$ and choose $S \subseteq E(H)$ with $|S| = k$ such that $H - S \to K$. Let $f\colon G \to H$ achieve $\mu(f) = \gamma(G, H)$. Define $T = \{e' \in E(G) : f(e') \in S\}$. For each H-edge in S, at most $\gamma(G, H)$ edges of G map to it, so $|T| \leq k \cdot \gamma(G, H)$. The map f restricted to $G - T$ lands in $H - S$ (since the only G-edges mapping to removed H-edges have been deleted), and $H - S \to K$, so $G - T \to K$. Hence $|G, K| \leq |T| \leq \gamma(G, H) \cdot |H, K|$. □

Lemma 3.5 (Two-sided bound via a common target)**.** *If $G \to H$ and $K \to H$ then*

$$|G, K| \le \gamma(G, H) \cdot |H, K| \qquad \text{and} \qquad |K, G| \le \gamma(K, H) \cdot |H, G|.$$

Consequently, $m(G, K) \le \max\big(\gamma(G, H) \cdot |H, K|,\ \gamma(K, H) \cdot |H, G|\big)$.

Proof. Both inequalities follow from Lemma 3.4, the first applied to the pair (G, H) with target K, the second to the pair (K, H) with target G. The bound on m follows by taking the max. □

Lemma 3.6 (Monotonicity in the target)**.** *If $H_1 \subseteq H_2$ (i.e. H_1 is a subgraph of H_2) and $G \to H_1$, then $\gamma(G, H_2) \le \gamma(G, H_1)$.*

Proof. The inclusion $\iota\colon H_1 \hookrightarrow H_2$ is an edge-injective homomorphism, so $\mu(\iota) = 1$ and $\gamma(H_1, H_2) \le 1$. Since $\gamma \ge 1$ always, $\gamma(H_1, H_2) = 1$. By Lemma 3.3, $\gamma(G, H_2) \le \gamma(G, H_1) \cdot \gamma(H_1, H_2) = \gamma(G, H_1)$. □

Proposition 3.2. *For $n \le m$, $\gamma(K_n, K_m) = 1$. For $n > m$, $\gamma(K_n, K_m) = +\infty$ (since $\chi(K_n) = n > m$ forbids any homomorphism).*

Proof. When $n \le m$ the inclusion $K_n \hookrightarrow K_m$ is edge-injective, so $\mu(\iota) = 1$ and $\gamma(K_n, K_m) \le 1$. Combined with Lemma 3.2, $\gamma(K_n, K_m) = 1$. □

3.1 Concentration for Regular Graphs

Theorem 3.1 (d-Regularity Formula)**.** *Let G be a d-regular graph on n vertices with $\chi(G) = 3$. Let*

$$M = \min\big\{\, \max(s_1, s_2, s_3) : s_1 + s_2 + s_3 = n,\ s_i \ge 0,\ s_1 \equiv s_2 \equiv s_3 \pmod 2 \,\big\}.$$

Then $\gamma(G, K_3) = \dfrac{d}{2} \cdot M$.

Proof. Let $f\colon G \to K_3$ be any homomorphism with colour classes A, B, C of sizes a, b, c (so $a + b + c = n$). Because G is d-regular, the number of edges between classes satisfies

$$e_{AB} = \tfrac{d}{2}(a + b - c), \quad e_{AC} = \tfrac{d}{2}(a + c - b), \quad e_{BC} = \tfrac{d}{2}(b + c - a).$$

(Each vertex in A contributes d edges total; all neighbours in A contribute 0 (no monochromatic edges), so $e_{AB} + e_{AC} = da$; the analogous identities for B and C give the three equations above.) Writing $s_1 = a + b - c$, $s_2 = a + c - b$, $s_3 = b + c - a$, the triple (s_1, s_2, s_3) is non-negative, sums to n, and all have the same parity (each $s_i \equiv a + b + c - 2x_i \equiv n \pmod 2$ for appropriate x_i). The max edge load under f is $\mu(f) = (d/2)\max(s_1, s_2, s_3)$.

Minimising over all achievable class distributions gives

$$\gamma(G, K_3) = \frac{d}{2} \cdot \min_{\substack{s_1+s_2+s_3=n,\ s_i \ge 0 \\ s_1 \equiv s_2 \equiv s_3 \pmod 2}} \max(s_1, s_2, s_3).$$

The minimum is M. (Any triple (a, b, c) of non-negative integers with $a + b + c = n$ is realised as an independent-set partition of any 3-chromatic d-regular graph provided no colour class is too large, which is guaranteed when G has no isolated vertices.) □

Remark 3.2. *The formula requires d to be even or the s_i to conspire so that $\frac{d}{2}s_i$ is an integer; this is automatic when n and d are both even, or when all s_i are even. The Petersen graph ($n = 10$, $d = 3$) requires same-parity triples summing to 10; all-even gives minimum max $= 4$ (e.g. $(4, 4, 2)$), so $\gamma = \frac{3}{2} \cdot 4 = 6$.*

4 The Symmetric Combination $m(G, H)$

Definition 4.1. $m(G, H) = \max\big(|G, H|,\ |H, G|\big)$.

The function m is clearly non-negative, satisfies $m(G, G) = 0$ (since $G \to G$), and is symmetric by definition. The following characterises when $m = 0$.

Proposition 4.1. *$m(G, H) = 0$ if and only if G and H are homomorphically equivalent (i.e. $G \leftrightarrow H$), which holds if and only if* $\mathrm{core}(G) \cong \mathrm{core}(H)$.

Proof. $m(G,H) = 0$ means $|G,H| = 0$ and $|H,G| = 0$, i.e. $G \to H$ and $H \to G$. This is precisely homomorphic equivalence, and two graphs are homomorphically equivalent iff they have isomorphic cores. □

Theorem 4.1. *m does* not *satisfy the triangle inequality in general.*

Proof. By Theorem 2.2 and the fact that $K_n \to K_m$ for all $n \le m$:

$$\begin{aligned} m(K_7, K_3) &= \max(|K_7, K_3|,\ |K_3, K_7|) = \max(5, 0) = 5, \\ m(K_7, K_5) &= \max(|K_7, K_5|,\ |K_5, K_7|) = \max(2, 0) = 2, \\ m(K_5, K_3) &= \max(|K_5, K_3|,\ |K_3, K_5|) = \max(2, 0) = 2. \end{aligned}$$

The triangle inequality would require $m(K_7, K_3) \le m(K_7, K_5) + m(K_5, K_3)$, i.e. $5 \le 4$. This fails. □

Remark 4.1. *Khoshkha [1] claims m is a pseudo-metric. Theorem 4.1 shows this is incorrect under Definition 2.1. A modified definition allowing edge additions as well as deletions, or using a different minimisation, might restore the triangle inequality; we leave this as an open question (Section 7, Question 3).*

5 Computational Results

We computed $|G,H|$ and $\gamma(G,H)$ using algorithms implemented in C++, Rust, and Python (source at https://github.com/rostam/NonHomomorphism):

- **Homomorphism search:** backtracking DFS with early constraint pruning (Rust, C++, and Python/NetworkX).
- **Non-homomorphism factor:** for $k = 0, 1, 2, \ldots$ test all $\binom{|E(G)|}{k}$ edge-removal subsets (C++ and Python).
- **Concentration parameter:** enumerate *all* homomorphisms and compute $\gamma = \min_f \mu(f)$ (C++ and Python).
- **Extended experiments (Section 6):** irregular source graphs, non-complete targets, and Kneser graphs (Python/NetworkX).

5.1 Cycles to K_3

Table 2: Non-homomorphism factor and concentration parameter for $C_n \to K_3$. The column $\lceil n/3 \rceil$ is the pigeonhole lower bound (Proposition 3.1); starred entries achieve it. Theorem 5.1 (verified by DP for $n \le 50$) gives the exact values in column γ.

C_n	$\lvert E\rvert$	$\lvert C_n, K_3\rvert$	$\gamma(C_n, K_3)$	$\lceil n/3 \rceil$
C_3	3	0	1^*	1
C_4	4	0	2^*	2
C_5	5	0	3	2
C_6	6	0	2^*	2
C_7	7	0	3^*	3
C_8	8	0	4	3
C_9	9	0	3^*	3
C_{10}	10	0	4^*	4
C_{11}	11	0	5	4

The data is consistent with the following exact formula, which holds for all $n \ge 3$ and is verified computationally for $n \le 50$.

Theorem 5.1. *For all $n \ge 3$,*

$$\gamma(C_n, K_3) = \begin{cases} \lceil n/3 \rceil & \text{if } n \not\equiv 2 \pmod 3, \\ \lceil n/3 \rceil + 1 & \text{if } n \equiv 2 \pmod 3. \end{cases}$$

Proof. We apply Theorem 3.1 with $d = 2$ (cycles are 2-regular). Write $n = 3q+r$ with $r \in \{0, 1, 2\}$. The parameter M is the minimum max of a same-parity non-negative triple summing to n.

Case $n \equiv 0 \pmod 3$ ($r = 0$, $n = 3q$). The triple (q, q, q) sums to $3q$ and all have the same parity. Its max is $q = n/3$; no smaller max is possible since the average is q. Hence $M = q$ and $\gamma = 1 \cdot q = n/3 = \lceil n/3 \rceil$.

Case $n \equiv 1 \pmod 3$ ($r = 1$, $n = 3q + 1$). The triple $(q + 1, q + 1, q - 1)$ has the same parity and sums to $3q + 1$; its max is $q + 1$. Any triple with max $\leq q$ sums to at most $3q < n$, so $M = q + 1 = \lceil n/3 \rceil$. Hence $\gamma = q + 1 = \lceil n/3 \rceil$.

Case $n \equiv 2 \pmod 3$ ($r = 2$, $n = 3q + 2$). Suppose there exists a same-parity triple summing to n with max $\leq q+1$. Then all three values are $\leq q+1$ and the sum is at most $3(q+1) = 3q+3$. But the sum must equal $3q+2$, so at least one value equals q. A triple with values in $\{q, q+1\}$ and sum $3q+2$ must have two values equal to $q+1$ and one to q. But q and $q+1$ have different parities, contradicting same-parity. Hence $M \geq q + 2$. The triple $(q + 2, q, q)$ achieves this (same parity: q and $q + 2$ are congruent mod 2), so $M = q + 2 = \lceil n/3 \rceil + 1$. Hence $\gamma = q + 2 = \lceil n/3 \rceil + 1$. □

Remark 5.1. *Theorem 5.1 follows from Theorem 3.1 by setting $d = 2$. The formula is* not *merely the pigeonhole bound $\lceil n/3 \rceil$: for $n \equiv 2 \pmod 3$ the parity obstruction raises γ by one above the pigeonhole value. Lemma 3.6 (monotonicity) is also consistent with Table 2: since $K_2 \subset K_3$, any homomorphism $C_n \to K_2$ (for even n) lifts to K_3 with no worse concentration.*

5.2 Complete Bipartite Graphs to K_3

Table 3: Non-homomorphism factor and γ for $K_{n,n} \to K_3$.

G	$\lvert E(G)\rvert$	$\lvert G, K_3\rvert$	$\gamma(G, K_3)$
$K_{1,1}$	1	0	1
$K_{2,2}$	4	0	2
$K_{3,3}$	9	0	6
$K_{4,4}$	16	0	8

Theorem 5.2. $\gamma(K_{n,n}, K_3) = n \cdot \lceil n/2 \rceil$.

Proof. Since $K_{n,n}$ is bipartite, $K_{n,n} \to K_2 \subset K_3$, so $|K_{n,n}, K_3| = 0$.

Let L and R be the two parts. In any homomorphism $f\colon K_{n,n} \to K_3$, the colour sets $f(L)$ and $f(R)$ must be disjoint: every vertex of L is adjacent to every vertex of R, and K_3 has no self-loops, so $f(\ell) \neq f(r)$ for all $\ell \in L$, $r \in R$. Since $|V(K_3)| = 3$, by the pigeonhole principle one part uses at most one colour and the other at most two.

Without loss of generality (by symmetry of $K_{n,n}$), suppose $f(R) \subseteq \{c\}$ and $f(L) \subseteq \{a, b\}$ with $\{a, b, c\} = V(K_3)$. If k vertices of L map to a and $(n-k)$ map to b, then the loads on $\{a, c\}$ and $\{b, c\}$ are nk and $n(n-k)$ respectively (each R-vertex contributes one edge to each L-vertex), and the load on $\{a, b\}$ is zero. The maximum is minimised at $k = \lfloor n/2 \rfloor$, giving $\max(n\lceil n/2 \rceil, n\lfloor n/2 \rfloor) = n\lceil n/2 \rceil$. Hence $\gamma(K_{n,n}, K_3) = n\lceil n/2 \rceil$. □

For $n = 1, 2, 3, 4$ this gives $1, 2, 6, 8$, matching Table 3.

5.3 The Petersen Graph

Table 4: Non-homomorphism factor and γ from the Petersen graph P to complete graphs. The pigeonhole bound is $\lceil 15/|E(H)| \rceil$.

H	$\lvert E(H)\rvert$	$\lvert P, H\rvert$	$\gamma(P, H)$	pigeonhole lb
K_2	1	3	$+\infty$	—
K_3	3	0	6	5
K_4	6	0	4	3
K_5	10	0	2	2

Observation 2.1 gives $|P, K_2| = 3$. Since $\chi(P) = 3$, $|P, K_m| = 0$ for all $m \geq 3$.

Observation 5.1. *$\gamma(P, K_3) = 6 > \gamma(P, K_4) = 4 > \gamma(P, K_5) = 2$. This is consistent with Lemma 3.6: since $K_3 \subset K_4 \subset K_5$, each inclusion is edge-injective and monotonicity gives $\gamma(P, K_3) \geq \gamma(P, K_4) \geq \gamma(P, K_5)$.*

Observation 5.2. *$\gamma(P, K_3) = 6$ strictly exceeds the pigeonhole lower bound $\lceil 15/3 \rceil = 5$. Theorem 3.1 explains this: with $n = 10$, $d = 3$, the optimal same-parity triple summing to 10 has minimum max $M = 4$ (e.g. $(4, 4, 2)$; all-odd is impossible since 10 is even), giving $\gamma = \frac{3}{2} \cdot 4 = 6$. The parity obstruction prevents $M = 3$ (which would require $(3, 4, 3)$, mixed parity).*

Observation 5.3. *The value $\gamma(P, K_5) = 2$ meets the pigeonhole bound $\lceil 15/10 \rceil = 2$. By Lemma 3.3 applied to $K_3 \to K_4 \to K_5$: $\gamma(P, K_5) \leq \gamma(P, K_3) \cdot \gamma(K_3, K_5) = 6 \cdot 1 = 6$ and $\gamma(P, K_4) \leq \gamma(P, K_3) \cdot \gamma(K_3, K_4) = 6 \cdot 1 = 6$. Both are consistent with (but do not imply) the computed values.*

5.4 Lemma Verification

Proposition 2.1(1): $K_3 \to K_4$, so $|P, K_3| \geq |P, K_4|$: $0 \geq 0$. ✓

Theorem 2.2 ($|K_{n+1}, K_n| = 1$): Verified computationally for $n = 2, 3, 4$. ✓

Lemma 3.3: $\gamma(P, K_4) = 4 \leq \gamma(P, K_3) \cdot \gamma(K_3, K_4) = 6 \cdot 1 = 6$. ✓

Lemma 3.4: $|P, K_2| = 3 \leq \gamma(P, K_3) \cdot |K_3, K_2| = 6 \cdot 1 = 6$. ✓

Theorem 4.1 (m fails triangle inequality): $m(K_7, K_3) = 5 > 4 = m(K_7, K_5) + m(K_5, K_3)$. ✓

6 Extended Computational Results

The preceding sections focused on regular source graphs (cycles, the Petersen graph) and complete target graphs. We now extend the computational scope to *irregular* source graphs, *non-complete* targets, and *Kneser graphs*.

6.1 Irregular Source Graphs

The d-regularity theorem (Theorem 3.1) applies only to regular graphs. To understand γ beyond this setting, we computed exact values for three families of irregular graphs, all with $\chi(G) = 3$.

Definition 6.1. *The* wheel graph *W_n has n vertices: a hub vertex adjacent to every vertex of a cycle C_{n-1}. It has $2(n-1)$ edges and $\chi(W_n) = 3$ when n is odd (even-length outer cycle), $\chi(W_n) = 4$ when n is even. The* friendship graph *F_k consists of k triangles sharing a common hub vertex ($2k + 1$ vertices, $3k$ edges, $\chi = 3$). The* book graph *B_k consists of k triangles sharing a common edge ($k + 2$ vertices, $2k + 1$ edges, $\chi = 3$).*

All three families are irregular: W_n has one vertex of degree $n - 1$ and $n - 1$ vertices of degree 3; F_k has hub degree $2k$ and leaf degree 2; B_k has two spine vertices of degree $k + 1$ and k page vertices of degree 2.

Table 5: Non-homomorphism factor and γ for wheel graphs to complete graphs. Wheels with $\chi = 4$ (even n) cannot map to K_3.

G	$\lvert V\rvert$	$\lvert E\rvert$	χ	H	$\lvert G, H\rvert$	$\gamma(G, H)$	$\lceil \lvert E\rvert / \lvert E(H)\rvert \rceil$
W_5	5	8	3	K_3	0	4	3
W_5	5	8	3	K_4	0	2	2
W_6	6	10	4	K_3	1	$+\infty$	—
W_6	6	10	4	K_4	0	3	2
W_7	7	12	3	K_3	0	6	4
W_7	7	12	3	K_4	0	2	2
W_8	8	14	4	K_3	1	$+\infty$	—
W_8	8	14	4	K_4	0	3	3
W_9	9	16	3	K_3	0	8	6
W_9	9	16	3	K_4	0	4	3

Tables 5–6 reveal several phenomena.

Table 6: Concentration parameter for friendship graphs F_k and book graphs B_k to K_3. All have $\chi = 3$ and $|G, K_3| = 0$. The column $\lceil |E|/3 \rceil$ is the pigeonhole lower bound.

G	$\|V\|$	$\|E\|$	$\|G, K_2\|$	$\gamma(G, K_3)$	$\lceil \|E\|/3 \rceil$
F_2	5	6	2	2	2
F_3	7	9	3	3	3
F_4	9	12	4	4	4
F_5	11	15	5	5	5
F_6	13	18	6	6	6
F_7	15	21	7	7	7
B_2	4	5	1	2	2
B_3	5	7	1	3	3
B_4	6	9	1	4	3
B_5	7	11	1	5	4
B_6	8	13	1	6	5
B_7	9	15	1	7	5

Observation 6.1. *$\gamma(F_k, K_3) = k$ and $|F_k, K_2| = k$ for all $k \geq 2$. The pigeonhole bound $\lceil 3k/3 \rceil = k$ is achieved exactly. This is because the hub vertex is adjacent to all $2k$ leaves; in any proper 3-colouring, the hub receives one colour and each triangle forces two distinct colours on its remaining pair. The optimal colouring assigns exactly one edge to each K_3-edge per triangle, so $\gamma = k$.*

Observation 6.2. *$\gamma(B_k, K_3) = k$ for all $k \geq 2$, but $|B_k, K_2| = 1$ for all k. The non-homomorphism factor is 1 because removing the spine edge $\{0, 1\}$ disconnects B_k into a star (bipartite), giving $B_k - \{0, 1\} \to K_2$. By contrast, γ grows linearly: any homomorphism to K_3 must map each page vertex to the third colour, piling all k page-edges onto a single K_3-edge. This demonstrates that γ and $|G, H|$ capture fundamentally different structural properties: $|G, H|$ measures global fragility (one removal suffices) while γ measures unavoidable local congestion.*

Observation 6.3. *For odd-n wheels W_n (with $\chi(W_n) = 3$), $\gamma(W_n, K_3) = 2(n-1)/3$ when $n \equiv 1 \pmod 3$ and exceeds the pigeonhole bound otherwise. For example, $\gamma(W_7, K_3) = 6$ whereas $\lceil 12/3 \rceil = 4$; the hub vertex creates a high-degree bottleneck that forces edge concentration.*

Table 7: γ for unbalanced complete multipartite graphs to K_3. These graphs are irregular and 3-chromatic.

G	$\|V\|$	$\|E\|$	$\gamma(G, K_3)$	$\gamma(G, K_4)$	$\lceil \|E\|/3 \rceil$
$K_{1,2,3}$	6	11	6	3	4
$K_{2,2,3}$	7	16	6	4	6
$K_{2,3,4}$	9	26	12	6	9

Observation 6.4. *For complete multipartite graphs $K_{a,b,c}$ ($a \leq b \leq c$), the concentration $\gamma(K_{a,b,c}, K_3)$ is determined by the "most loaded" K_3-edge, which corresponds to the two largest parts. For $K_{2,3,4}$: the optimal 3-colouring assigns each part to a distinct colour, and the most loaded edge connects the parts of size 3 and 4, giving $\gamma = 3 \times 4 = 12$. More generally, $\gamma(K_{a,b,c}, K_3) = \max(ab, ac, bc)$.*

6.2 Non-Complete Target Graphs

All results above use complete graphs K_m as targets. We now consider $\gamma(G, H)$ where H is an odd cycle or the Petersen graph.

Observation 6.5. *Table 8 reveals that $C_n \to C_5$ if and only if n is even or n is an odd multiple of 5. More precisely, $C_n \to C_m$ (for odd m) if and only if n is even, or m divides n. The concentration $\gamma(C_n, C_5)$ achieves the pigeonhole bound $\lceil n/5 \rceil$ when $5 \mid n$, and exceeds it (typically by 1) otherwise — analogous to the parity obstruction seen for $\gamma(C_n, K_3)$ but arising from the circular structure of C_5.*

Observation 6.6. *When the target H is the Petersen graph (which has 15 edges and girth 5), the concentration values are remarkably low — often $\gamma(C_n, P) = 1$. This reflects the Petersen*

Table 8: $\gamma(C_n, C_5)$ for cycles mapping to the 5-cycle. C_5 has 5 edges. Starred entries achieve the pigeonhole lower bound $\lceil n/5 \rceil$. Note: $C_3 \not\to C_5$ (since $\chi(C_3) = 3$ but C_5 has odd girth 5 and cannot receive a triangle homomorphically).

C_n	$\gamma(C_n, C_5)$	$\lceil n/5 \rceil$	achieves bound?
C_3	$+\infty$	—	—
C_4	2	1	no
C_5	1^*	1	yes
C_6	2^*	2	yes
C_7	3	2	no
C_8	2^*	2	yes
C_9	3	2	no
C_{10}	2^*	2	yes
C_{11}	3^*	3	yes
C_{12}	4	3	no
C_{13}	3^*	3	yes
C_{14}	4	3	no
C_{15}	3^*	3	yes

Table 9: $\gamma(C_n, C_7)$ for cycles mapping to the 7-cycle. $C_n \to C_7$ iff n is even or $7 \mid n$.

C_n	$\gamma(C_n, C_7)$	$\lceil n/7 \rceil$	achieves bound?
C_3	$+\infty$	—	—
C_4	2	1	no
C_5	$+\infty$	—	—
C_6	2	1	no
C_7	1^*	1	yes
C_8	2	2	no
C_9	3	2	no
C_{10}	2	2	no
C_{11}	3	2	no
C_{12}	2	2	no
C_{13}	3	2	no
C_{14}	2^*	2	yes
C_{15}	3	3	no

graph's high symmetry (vertex- and edge-transitive) and large number of edges, which allow homomorphisms to distribute edge loads nearly uniformly. Note that $C_3 \not\to P$ because the Petersen graph is triangle-free ($\mathrm{girth}(P) = 5$*).*

6.3 Kneser Graphs

Kneser graphs form a central family in homomorphism theory. The Kneser graph $K(n, r)$ has vertices corresponding to the r-element subsets of $\{1, \ldots, n\}$, with two vertices adjacent iff the subsets are disjoint. Its chromatic number is $\chi(K(n, r)) = n - 2r + 2$ (Lovász [4]).

Observation 6.7. *The Kneser graph $K(6, 2)$ has 15 vertices, 45 edges, and $\chi(K(6, 2)) = 4$. Since $\chi(K(6, 2)) > 3$, we have $|K(6, 2), K_3| > 0$; the computed value $|K(6, 2), K_3| = 3$ shows that three edge removals suffice to make $K(6, 2)$ 3-colourable. The value $\gamma(K(6, 2), K_4) = 9$ exceeds the pigeonhole bound $\lceil 45/6 \rceil = 8$, indicating that even K_4 with 6 target edges cannot achieve a perfectly balanced load distribution.*

6.4 Extended Lemma Verification

All extended results are consistent with the theoretical framework.

Multiplicativity (Lemma 3.3): $\gamma(F_4, K_4) = 3 \leq \gamma(F_4, K_3) \cdot \gamma(K_3, K_4) = 4 \cdot 1 = 4$. ✓
$\gamma(W_7, K_4) = 2 \leq \gamma(W_7, K_3) \cdot \gamma(K_3, K_4) = 6 \cdot 1 = 6$. ✓

NHF Bound (Lemma 3.4): $|F_4, K_2| = 4 \leq \gamma(F_4, K_3) \cdot |K_3, K_2| = 4 \cdot 1 = 4$. ✓
$|W_7, K_2| = 3 \leq \gamma(W_7, K_3) \cdot |K_3, K_2| = 6 \cdot 1 = 6$. ✓

Table 10: $\gamma(C_n, P)$ for cycles mapping to the Petersen graph (15 edges). Every C_n with $n \geq 4$ maps to P since P contains even cycles (as a non-bipartite graph with girth 5) and 5-cycles, but $C_3 \not\to P$ since P is triangle-free.

C_n	$\gamma(C_n, P)$	$\lceil n/15 \rceil$	achieves bound?
C_3	$+\infty$	—	—
C_4	2	1	no
C_5	1^*	1	yes
C_6	1^*	1	yes
C_7	2	1	no
C_8	1^*	1	yes
C_9	1^*	1	yes
C_{10}	2	1	no
C_{11}	2	1	no
C_{12}	2	1	no

Table 11: Kneser graphs: γ and $|G, H|$ values. $K(5, 2)$ is the Petersen graph. $K(6, 2)$ has $\chi = 4$ and is the smallest Kneser graph requiring more than 3 colours.

G	$\lvert V\rvert$	$\lvert E\rvert$	χ	H	$\lvert G, H\rvert$	$\gamma(G, H)$
$K(5, 2)$	10	15	3	K_3	0	6
$K(5, 2)$	10	15	3	K_4	0	4
$K(5, 2)$	10	15	3	K_5	0	2
$K(6, 2)$	15	45	4	K_3	3	$+\infty$
$K(6, 2)$	15	45	4	K_4	0	9

Monotonicity (Lemma 3.6): $\gamma(F_4, K_4) = 3 \leq \gamma(F_4, K_3) = 4$. ✓
$\gamma(W_7, K_4) = 2 \leq \gamma(W_7, K_3) = 6$. ✓
$\gamma(B_5, K_4) = 3 \leq \gamma(B_5, K_3) = 5$. ✓

Pigeonhole (Proposition 3.1): The bound $\gamma(G, H) \geq \lceil |E(G)|/|E(H)| \rceil$ is tight for $F_k \to K_3$ (all k) and for $C_n \to C_5$ when $5 \mid n$, but is *not* achieved for $W_7 \to K_3$ ($6 > 4$), $K_{2,3,4} \to K_3$ ($12 > 9$), or $K(6, 2) \to K_4$ ($9 > 8$).

7 Open Questions

1. **Extend the d-regularity formula to general H.** Theorem 3.1 gives $\gamma(G, K_3)$ exactly for d-regular G with $\chi(G) = 3$. Is there an analogous formula for $\gamma(G, H)$ when H is an arbitrary k-chromatic graph? The argument uses the fact that K_3 has a unique edge type (all 3 edges are equivalent under automorphisms); for H with multiple edge orbits the formula would need to track the load on each orbit separately.

2. **When does $\gamma(G, H)$ meet the ratio $|E(G)|/|E(H)|$?** For d-regular G with $\chi(G) = 3$, Theorem 3.1 shows that $\gamma(G, K_3)$ meets the pigeonhole bound $\lceil |E(G)|/|E(K_3)| \rceil$ exactly when $n \equiv 0 \pmod 3$ (equal class sizes achievable), and exceeds it by $d/2$ when $n \equiv 2 \pmod 3$ (parity obstruction). For the Petersen graph ($n = 10$, $d = 3$, $n \equiv 1 \pmod 3$) we get $\gamma(P, K_3) = 6 > 5 = \lceil 15/3 \rceil$. Does $\gamma(G, H)$ always equal the ratio $|E(G)|/|E(H)|$ when H is "large enough" relative to G? (Computations show $\gamma(P, K_5) = 2$ meets the pigeonhole bound $\lceil 15/10 \rceil = 2$.)

3. **Can m be repaired into a metric?** Theorem 4.1 shows $m(G, H) = \max(|G, H|, |H, G|)$ does not satisfy the triangle inequality. Two variants worth investigating are: (a) $d(G, H) = \min\{k : \exists G' \text{ with } |E(G) \triangle E(G')| = k \text{ and } G' \to H\}$ (edge-edit distance to a subgraph mapping to H); (b) $m'(G, H) = \max(|G, H|, |H, G|)$ restricted to graphs with the same vertex count. Do either of these satisfy a triangle inequality?

4. **γ and the fractional chromatic number.** For vertex-transitive G, $\chi_f(G) = |V(G)|/\alpha(G)$. The Petersen graph has $\chi_f(P) = 5/2$. Is $\gamma(G, K_m)$ related to $\chi_f(G)$? For example, does $\gamma(G, K_m) \geq |E(G)|/(m\lfloor |V(G)|/m \rfloor)$ hold for vertex-transitive G? For P and $m = 3$: $15/(3 \cdot$

3) = 5/3, which gives $\lceil 5/3 \rceil = 2$, weaker than the actual value 6. A tighter bound using χ_f may explain why $\gamma(P, K_3) = 6$ instead of 5.

5. **Characterise when $|G, K| \leq |G, H| + |H, K|$.** The triangle inequality fails for m, but the sub-additivity of $|G, \cdot|$ might hold under additional conditions. For complete graphs, $|K_n, K_m| + |K_m, K_\ell| = |K_n, K_\ell|$ when n, m, ℓ are in arithmetic progression (e.g. K_4, K_3, K_2: $1 + 1 = 2\checkmark$), but fails when the gap grows faster (e.g. K_7, K_5, K_3: $2 + 2 < 5$). A systematic search finds 108 triples (K_a, K_b, K_c) with $a, b, c \leq 10$ where $m(K_a, K_c) > m(K_a, K_b) + m(K_b, K_c)$; the excess is largest when $b \approx \sqrt{ac}$ (the "geometric middle"), which maximises the Turán discrepancy $t(a, b) - t(a, c) > |K_b, K_c|$. What characterises the triples (G, H, K) for which sub-additivity holds?

References

References

[1] K. Khoshkha, *Nonhomomorphism Factors*, M.Sc. thesis, Sharif University of Technology, 2005.

[2] A. Daneshgar and H. Hajiabolhassan, "Circular colouring and algebraic no-homomorphism theorems," *European Journal of Combinatorics*, vol. 28, no. 6, pp. 1843–1853, 2007.

[3] A. Daneshgar and H. Hajiabolhassan, "Graph homomorphisms through random walks," *Journal of Graph Theory*, vol. 44, pp. 15–38, 2003.

[4] P. Hell and J. Nešetřil, *Graphs and Homomorphisms*, Oxford University Press, 2004.

[5] Y. Shitov, "Counterexamples to Hedetniemi's conjecture," *Annals of Mathematics*, vol. 190, no. 2, pp. 663–667, 2019.